UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5D - Cognition and Human Information Processing I

EIN 3241: Human Factors and Ergonomics
Spring 2026

March 2, 2026

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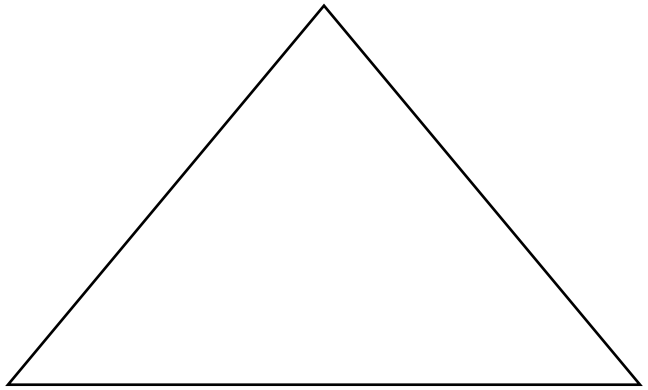
Admin Issues

- Lab 4 due Friday
- Midterm in a week!

Cognitive System Triad (Woods & Roth, 1988)

Agents
(humans and machines – joint cognitive system)

- Human information processing
  - Perceptual characteristics
  - Sources of error



World
(to be acted upon; often characterized by risk, dynamism, complexity, uncertainty)

Representation
(through which agents experience, and act on, the world)

- Visual form
- Directability
- Observability

3

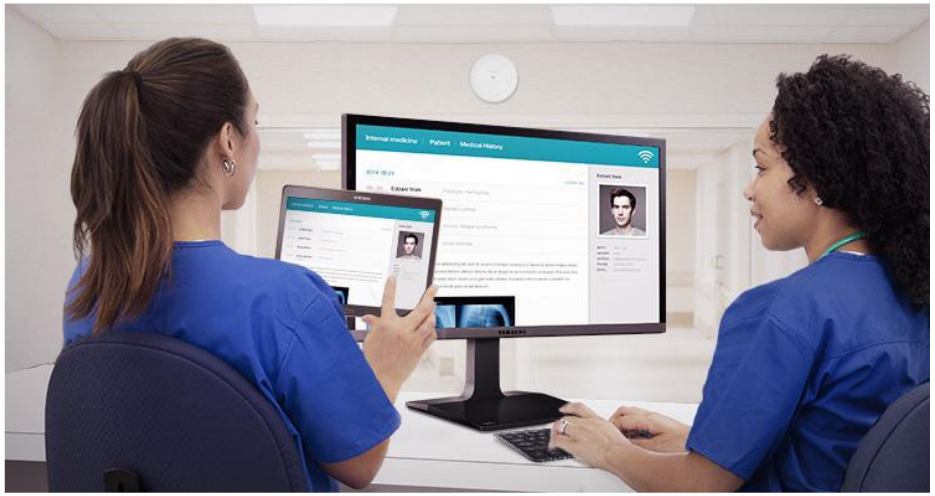
Cognitive Ergonomics

- Cognitive ergonomics: concerned with the mental processes of information in occupational settings, in order to optimize the interactions among humans and other elements of a system.
- Growing need for cognitive ergonomics due to the increase in automation and service jobs.

Applications of Cognitive Ergonomics:

Designing a cockpit

Designing Electronic Medical Records

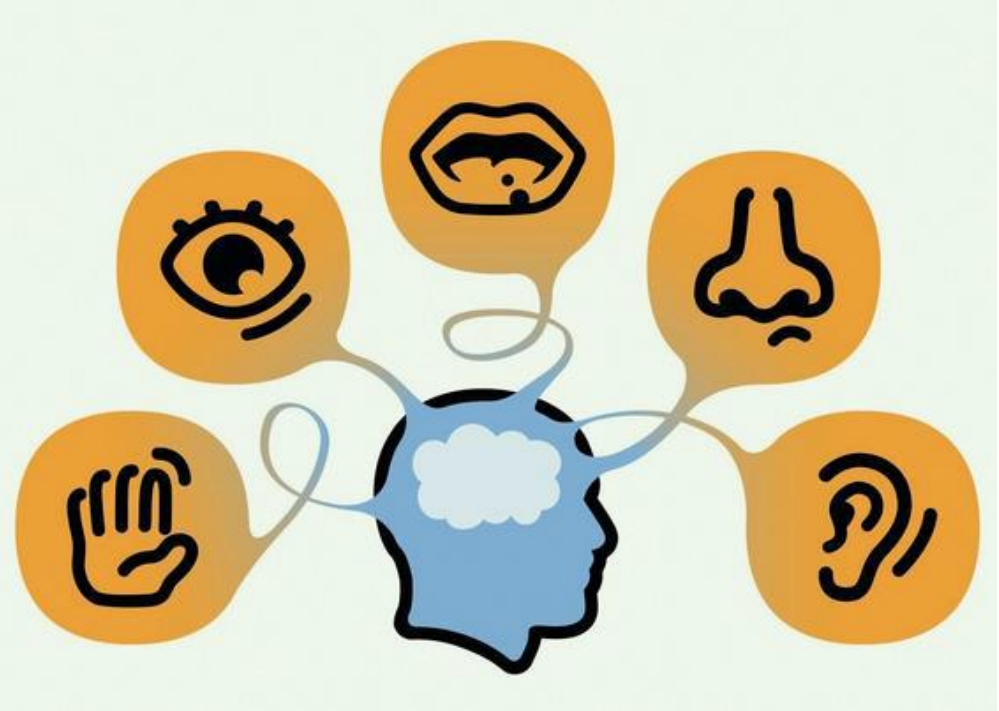


Designing Phone Interfaces

Designing a nuclear power plant control system

Information

- Information: interpreted data acquired through any of the five senses.
- The more sense modalities used in presenting one piece of information, the more likely the stimuli will be recognized and remembered.
- Most information is acquired from sight and hearing (95% 99%).



Cognition and decision making

- Three stages of human information processing:
  - Perception of information (also see previous lectures on signal detection)
  - Central processing of information
  - Responding to information

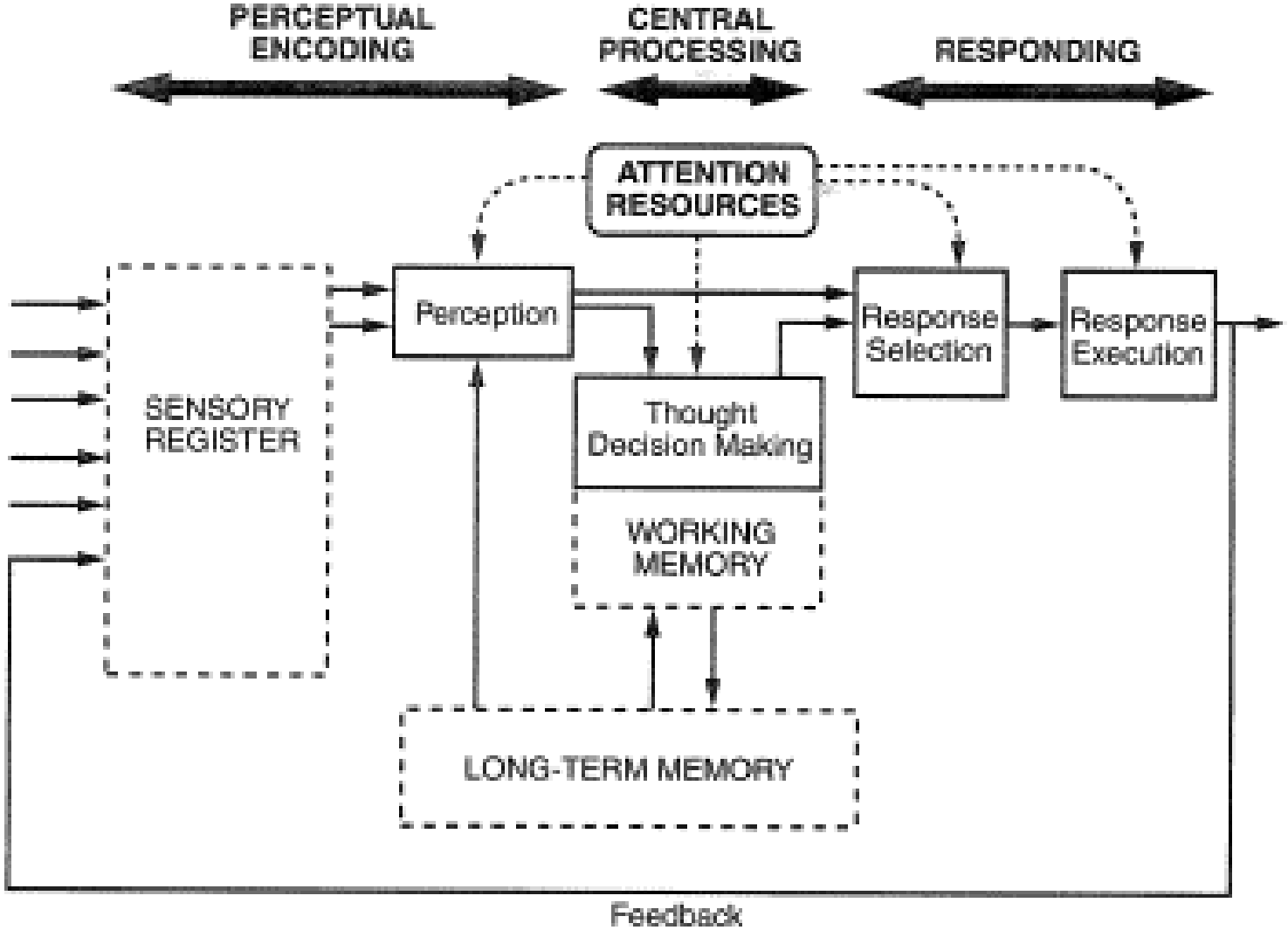
Cognition

Decision making

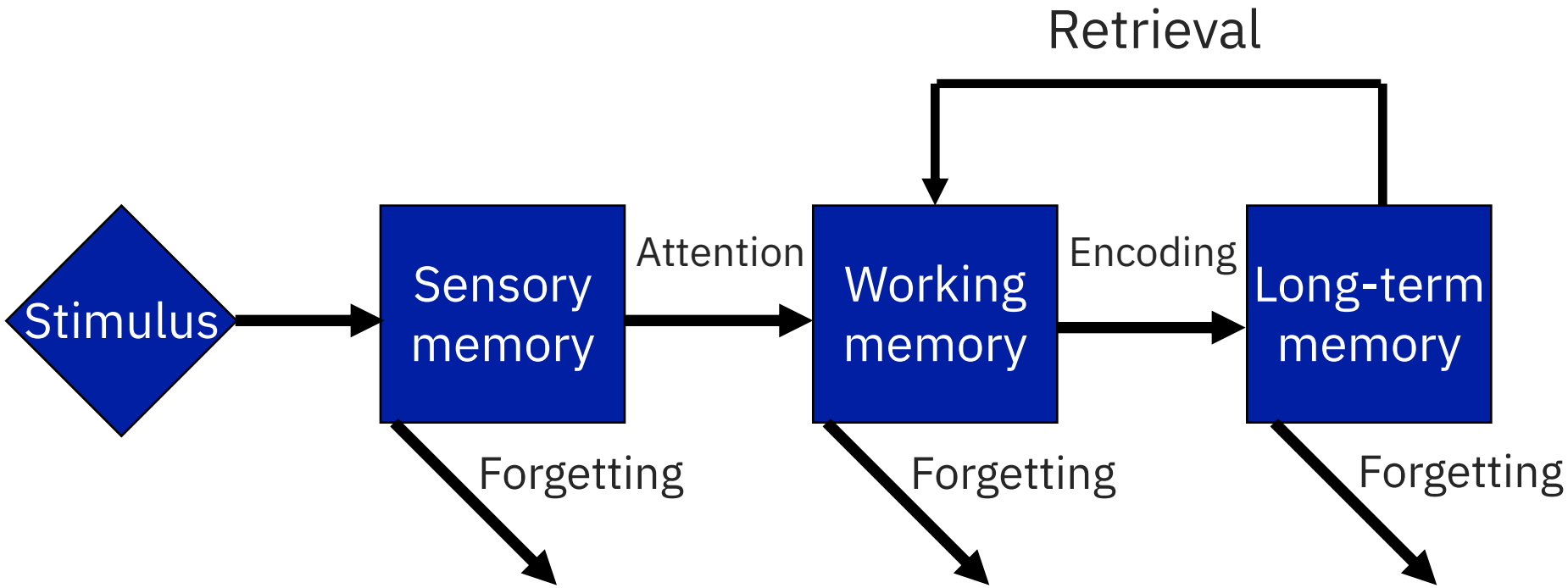
- Cognition is “top-down” process:
  - Uses existing knowledge (stores) to interpret current experiences (of environment)
  - Goals and mental models apply

- Decision making is based on cognition and is also “top down” process with biases based on experience and prior task performance

Wickens et al.'s Model of Human Information Processing



Simple “pipeline” model of HIP and memory



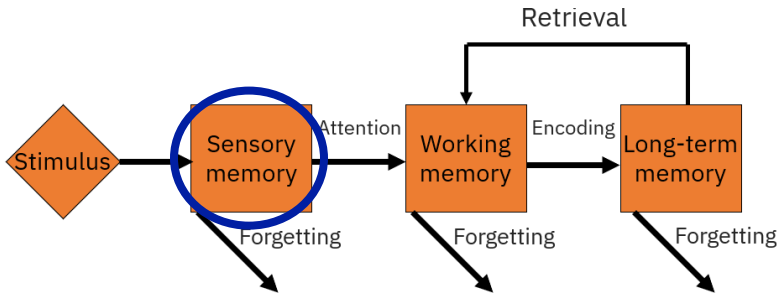
Information processing and Memory

- Basic assumptions of any model:
  - Information is transformed (altered in code) from one stage (or "construct") to another
  - Each stage takes some time for processing
- Processes related to memory:
  - **Encoding** – Acquiring and transforming information (process)
  - **Storage** – Retention of information in brain (process and construct (WM, LTM))
  - **Retrieval** – Recall of information from memory at later time (process)
  - **Forgetting** – Failure to recall material that was remembered (due to lack of attention; process)
- Three aspects of memory:
  - **Function** – What is done with stored info.
  - **Capacity** – How much info can be stored.
  - **Duration** – How long can info be retained.

  (To be discussed for each type…)

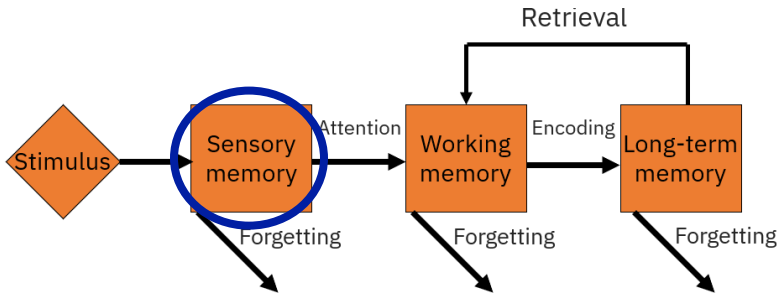
- Encoding process dictates whether something is remembered or forgotten.
- Interruptions in encoding and similarity of stimuli compromise encoding.

Sensory memory



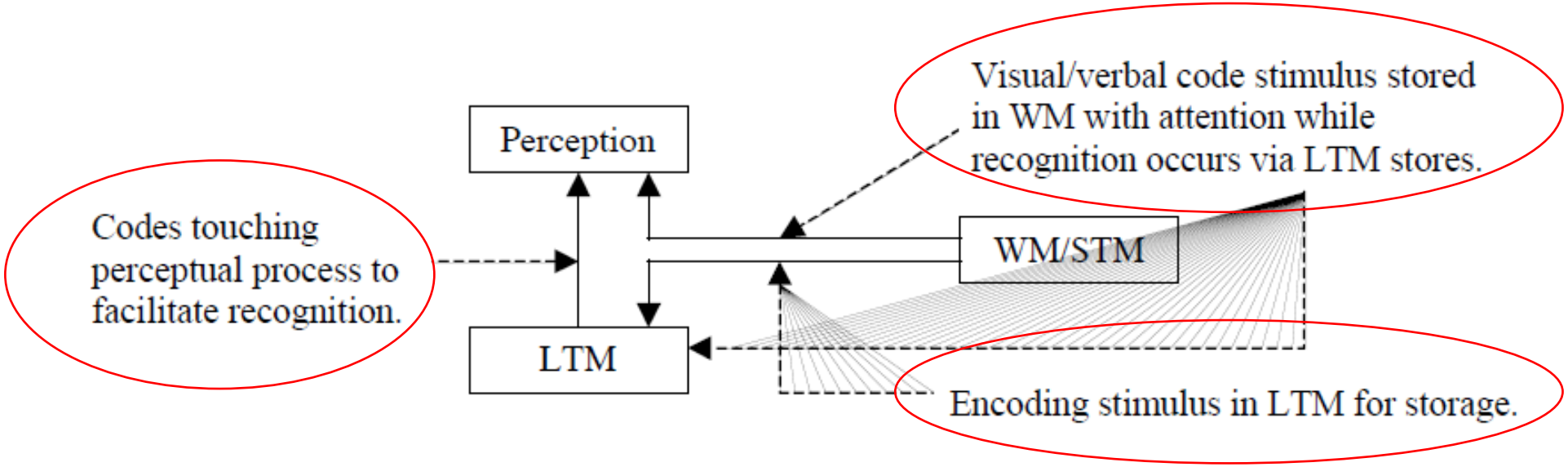
- Stores stimuli detected by sensory receptors:
  - Sensory memory forms automatically, without attention or interpretation
  - However, attention is needed to transfer information to working memory to support perceptual process
- Large capacity - Can hold many items at once, but information is “unprocessed” (no encoding) based on recognition through use of LTM.
- Short-duration buffer with rapid decay
- Three types (with varying duration):
  - Iconic memory:
    - Visual stimuli
    - Usually lasts about 0.3 seconds (Efron, 1970)
  - Echoic memory
    - Sensory memory for auditory input
    - Lasts only 2 to 3 seconds
  - Kinesthetic
    - Tactile stimuli
    - Lasts for several seconds after contact

Perception

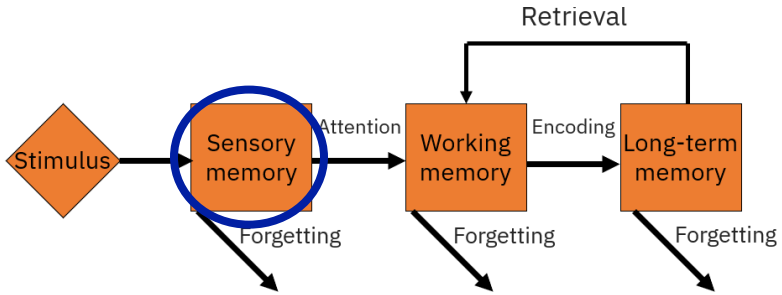


- Recognition:
  - Assign meaning to stimuli captured by senses
  - Compare incoming stimulus with stored patterns in LTM for recognition

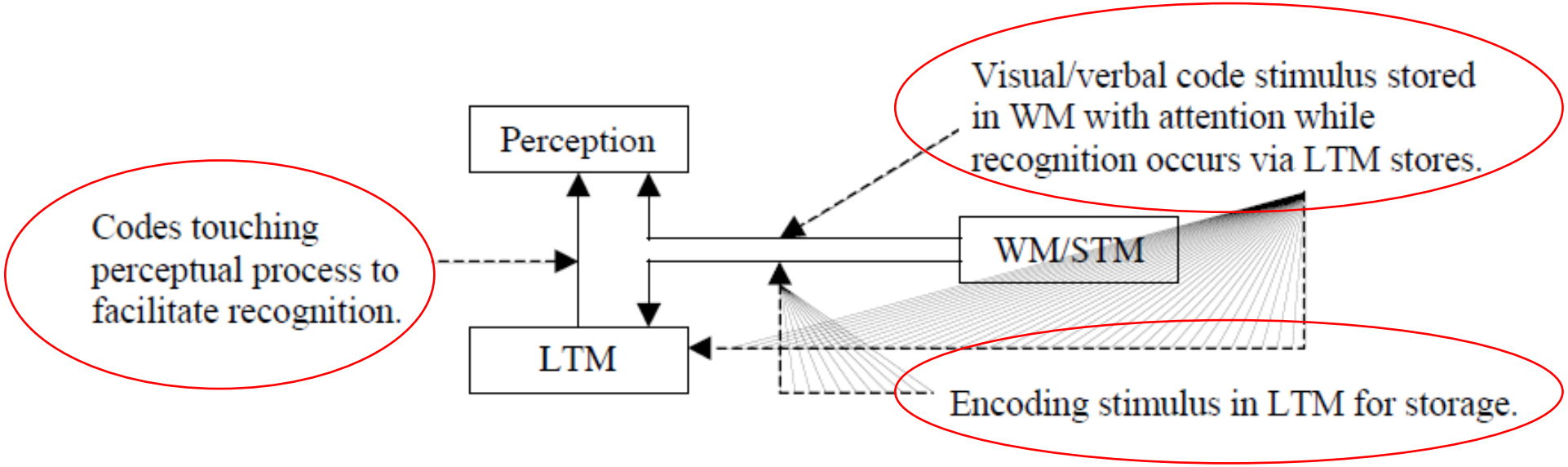
- Attention to stimulus leads to feature identification via LTM
- Features then temporarily stored in WM/STM
- Once feature set is complete, WM content can be encoded for LTM



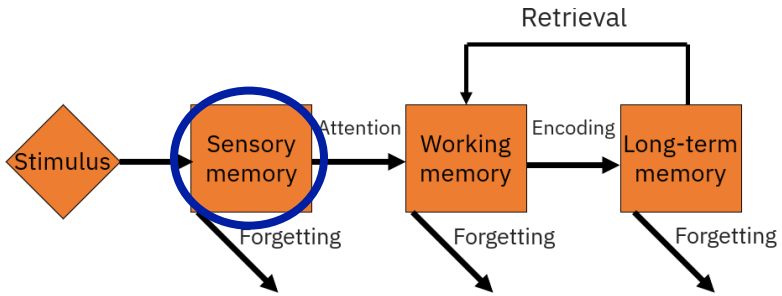
Perception



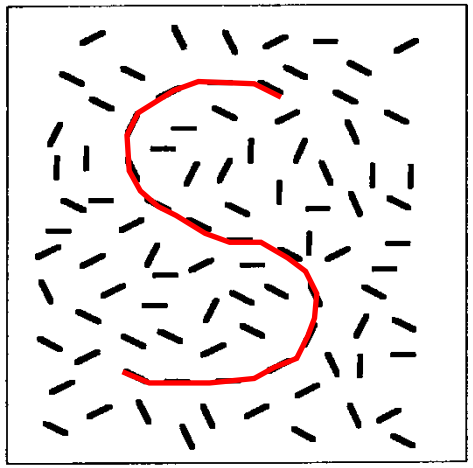
- Three processes can be identified here…
  - Bottom-up feature analysis
  - "Unitization"
  - Top-down processing



Bottom-up processing

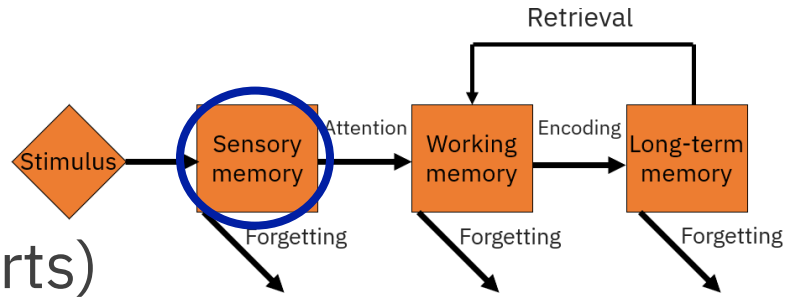


- Do you see a letter imbedded in the lines?



- Recognition of patterns in stimuli based on low-level features:
  - Attempt to combine component features
  - Match observed patterns to <u>stored patterns in LTM</u>
  - Decide which stored pattern is best match
- Requires use of LTM for prior knowledge and WM/STM for temporary storage of perceived patterns.

Unitization



- When stimulus is perceived as “whole” (vs. sum of parts)
  - As familiarity with perceptual pattern increases, stimulus processing transform from feature analysis to global or holistic processing (Gestalt of image)
  - Allows humans to read familiar words and phrases rapidly and overlook typographical errors
  - Gestalt is a theory of visual perception with <u>principles</u> identifying specific forms of perceptual processing
    - Similarity occurs when there are similar features of co-located but different stimuli
    - Continuation occurs when the eye moves through one object to complete another
    - Closure occurs when objects are incomplete or space is not enclosed
    - Anomaly detection occurs when one part is dissimilar from others comprising the “whole”

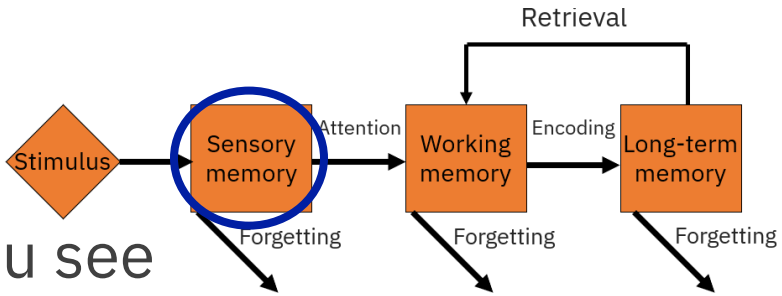
Similarity

Continuation

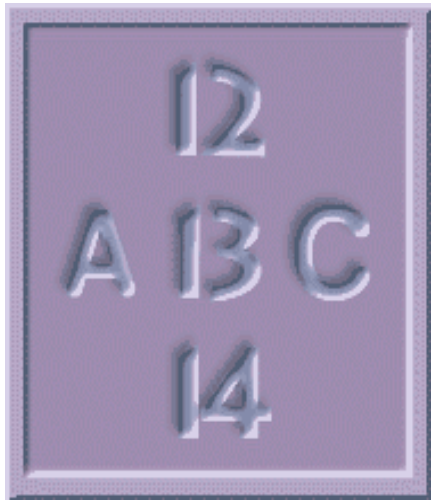
Closure

Anomaly

Top-down processing



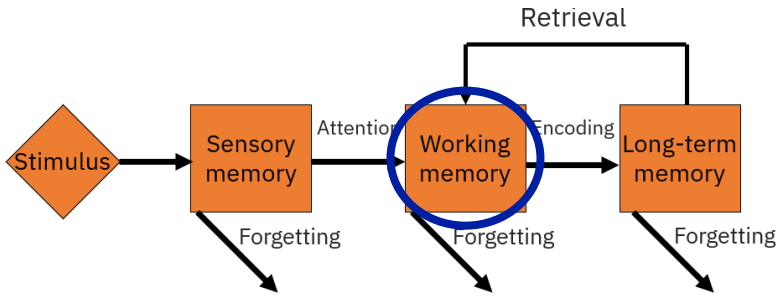
- Prior experiences and expectations determine what you see
  - e.g., strong familiarity with image of Lincoln leads to identification of face; absence of familiarity lead to identification of figure of woman at center of image
  - e.g., individual preference for verbal encoding leads to identification of letter "B"; preference for numerical encoding leads to identification of number “13”



Top-down and bottom-up processing implications for interface design

- Maximize bottom-up processing
  - Make features easy to identify using size and contrast
  - Highlight patterns or feature cueing pattern (e.g., compass pointer)
- Maximize top-down processing
  - Use words, as goals are typically verbally encoded in WM
  - Minimize number of words that need to be recognized
  - Provide contextual information
    - e.g., grouping of icons according to function
- Maximize automaticity and unitization:
  - Use familiar perceptual representations (those encountered frequently and stored in LTM)
  - e.g., meaningful icons (clock face, calculator functions), familiar fonts and mixed text, and words rather than abbreviations

Working memory (WM)

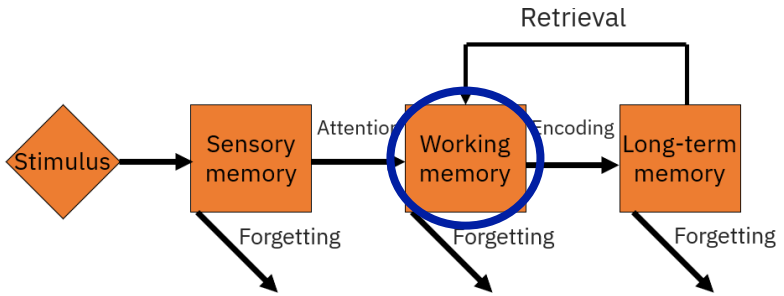


- Role: Keeps information active while in use
- Not a synonym for short term memory!
  - Refers to short term store needed for **certain mental tasks**
  - Defined not in terms of duration, but rather in terms of **purpose**
- Short-term memory (STM) usually refers to the temporary storage of a small amount of information for a brief period (seconds to minutes) **without active manipulation**.
  - Example: remembering a phone number just long enough to dial it.
  - It's more about holding information.
- Working memory (WM) includes short-term storage **plus the active processing and manipulation** of that information to achieve a goal
  - Example: remembering a number while also calculating its sum with another number. It's more about holding and working with information.

Working memory: Design implications

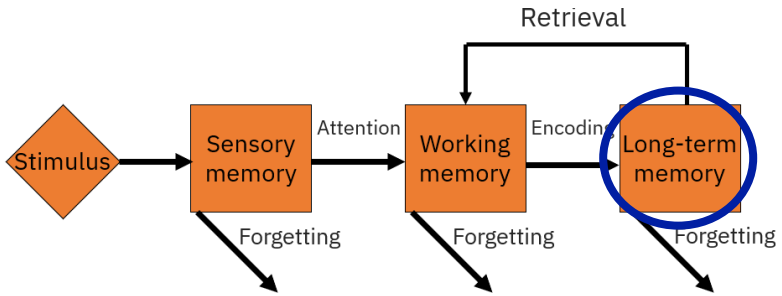
- Minimize WM load (time x chunks)
- Use redundant sources (text + speech)
- Reminders for sequential tasks (e.g., checklists)
- Exploit chunking
  - Size of item clusters 2376100 OR 237-6100?
  - Use meaningful sequences
  - Letters rather than numbers (1-800-GET-HELP)
  - Separate numbers and letters
- Minimize confusion
- Use different memory codes (spatial and verbal)
- Order text/instructions carefully

Working memory (WM)



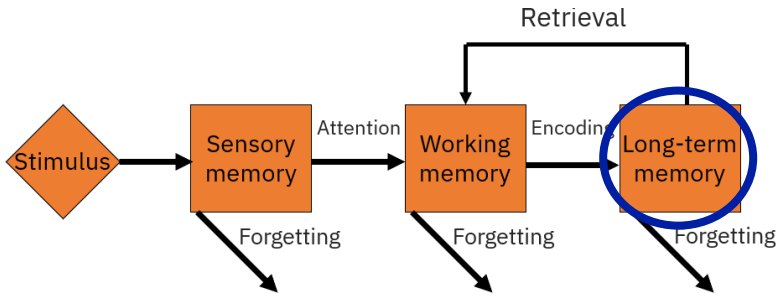
- Miller's chunking theory: 7±2 "chunks" of information is the limit of the working memory.
- A chunk is a meaningful unit of information created by grouping smaller bits together using prior knowledge.
  - Example: Remembering 1 9 4 5 as "1945" (a year) instead of four separate digits.
  - Example: In language, the word "dog" is a single chunk rather than three separate letters.
- Chunking works because WM leverages LTM to combine related elements into a single, meaningful unit, effectively expanding what you can store.
- Experts in a domain can remember more complex patterns because they have richer, more meaningful chunks in long-term memory (e.g., a chess master seeing whole board configurations rather than isolated pieces).
- WM is often described as a bottleneck for cognition
  - Only a few chunks can be actively maintained and manipulated at once.
- If information in WM (or STM) is not actively rehearsed, it fades within about 15–30 seconds.
- If you have more chunks to hold, each chunk gets less rehearsal time, making decay more likely.
  - The total time before an item is lost depends on how quickly you cycle through the items during rehearsal.

Long-term memory (LTM)



- Role: Keeps event information available for recall over long periods of time
- Key characteristics:
  - A storage of information in the brain for a long or an indefinite period.
  - Simplifies working memory processing
  - Establishes expectations about the world
  - Capacity and duration are unlimited.
- Process:
  - Item strength = f { frequency, recency, importance }
  - Forgetting (weak strength, few associations, interfering associations)
- Short term memories can become long term memories through rehearsal and/or encoding information semantically (i.e. based on meaningful association

Long-term memory

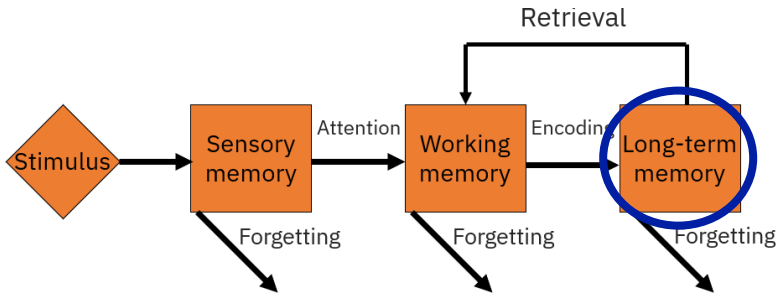


- **Semantic or declarative memory:** memory of things (general knowledge) – *"knowing that"*
- **Procedural memory:** memory of how to do things – *"knowing how"*
- **Episodic memory:** memory for specific events

Why might our memories of past events be biased?

- Encoding
- Storage
- Retrieval

Long-term memory



Guidelines:

- Don’t violate people’s mental framework, i.e., design according to human expectations.
- Place most important information at the beginning or end.
- Relate new information to previous information (to integrate into existing frameworks).
- Appeal to the emotional component of memory by using humor and emotion.
- Storytelling: helps provide a framework and also appeals to emotion.

LTM: Design implications

- Encourage regular use (frequency, recency)
- Standardize (frequency, recency)
- Memory aids (supplement LTM)
- Carefully designed information (limit LTM dependence, support associations)
- Reminders can be used to avoid failures of prospective memory

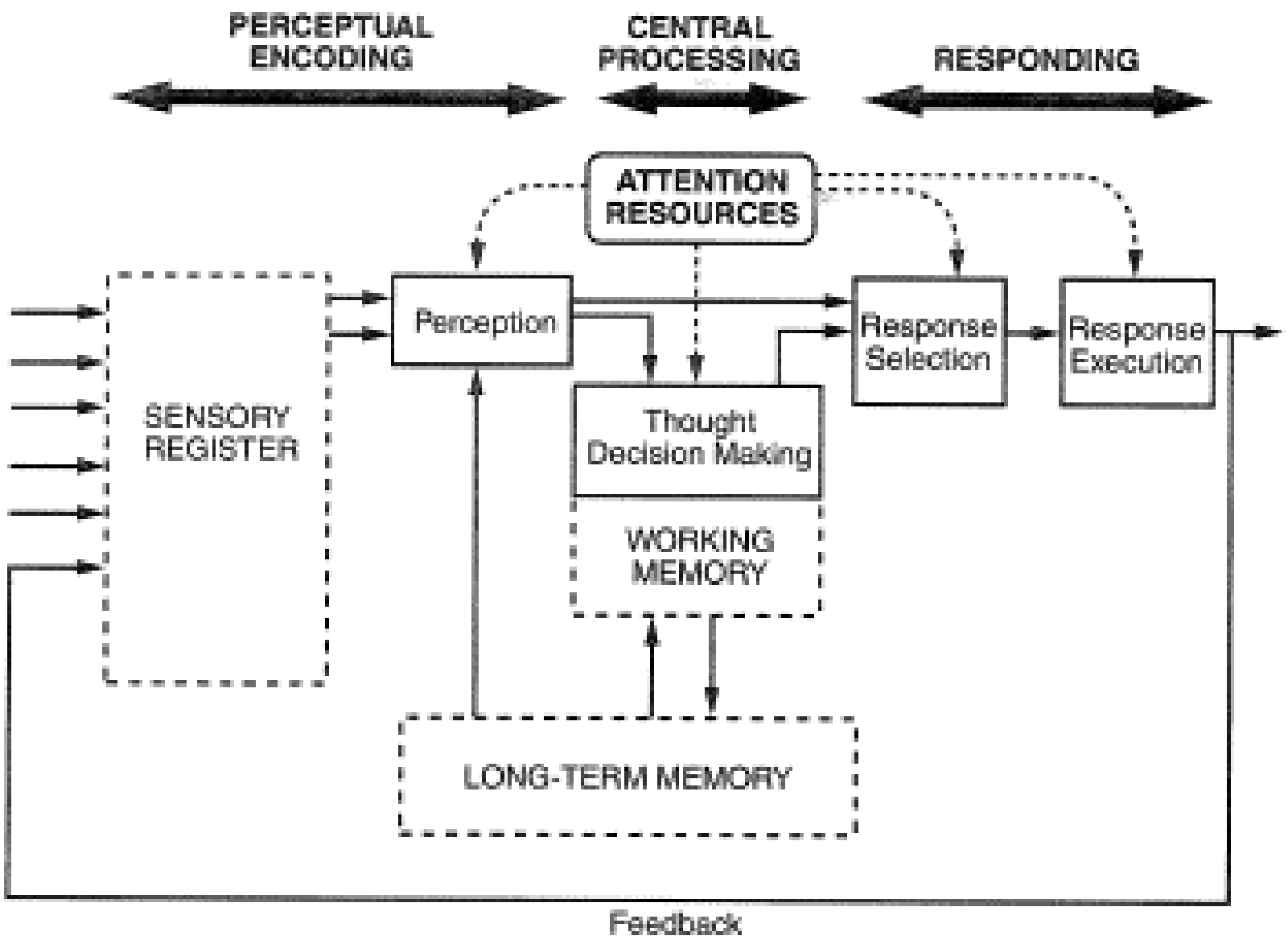
Organization of Information in LTM

- **Semantic network:** Connected pieces of semantic information (e.g., concept maps)
- **Schema and scripts:** Knowledge structure about a particular topic. (Scripts – schema that is a sequence of activities)
- **Mental models:** Schemas of dynamic systems (e.g., equipment, technologies, or tools)
- **Cognitive maps:** Mental representations of spatial information (e.g., layout of a room)

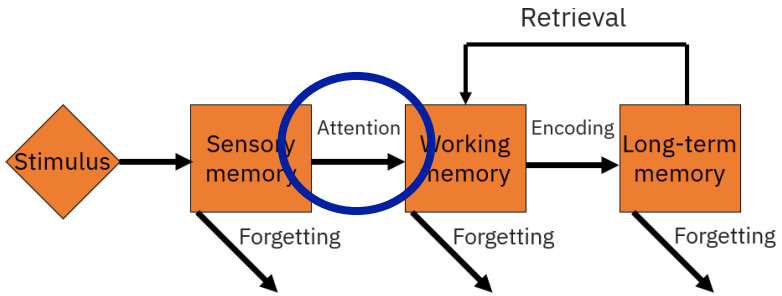
Attention

Attention as a selected information process. “Filter”

Attention as a limited resource. “Fuel”



Attention



- Attention is the process of selecting which information to process.
  - Attention as selective information processing - Analogies include:
    - Channel switching
    - Shining a flashlight in a dark attic

- Attention is required to support most cognitive activities (not LTM or sensory registers)
  - Attention as a limited resource.

Types of Attention

- Selective/Focused Attention
  - Maintain attention on one or a few sources of information and not be distracted by other sources
- Divided Attention
  - Required to do more than one task at the same time (e.g., cocktail party effect)
- Sustained Attention
  - The ability of observers to maintain attention and remain alert to stimuli over prolonged periods of time (i.e., vigilance)

Selective Attention: Design Implications

- Role: Ability to focus on a specific set of stimuli.
  - Selective attention does not guarantee perception but is a necessary condition.
- High value information sources should require minimal effort to attend
- Important, unexpected or rare events should be conveyed in a salient manner

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LEADING THE CHARGE, CHARGING AHEAD